

VALEN LI

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EDUCATION

Julia R. Masterman High School

GPA: 4.53/4.0

Robotics Team (Co-Founder & Lead Programmer) | Chess Team (Co-Captain) | AI Club (President)

Philadelphia, PA

Sep 2018 – June 2026

Community College Of Philadelphia

Courses: MATH 163: Discrete Mathematics I — Math 263 Discrete Mathematics II

In-Person

Sep 2025 – Jun 2025

The Wharton School of the University of Pennsylvania

3-week residential program at UPenn called: The Leadership in the Business World

Residential

Jul 2025 – Aug 2025

Oberlin College

Pioneer Research Program (Under Full-Scholarship and given Four College Credits)

Remote

Feb 2025 – September 2025

EXPERIENCE

Julia R. Masterman Secondary School

Technology Teacher Leader Aide / IT

Philadelphia, PA

Sep 2025 – Present

- Managed Chromebook deployment and configuration for students
- Provided technical support and assistance to teachers and students, resolving hardware and software issues
- Assisted teachers with laptop repairs, inventory indexing, cable management, and grading papers
- Redesigned the school [website](#)

Wharton's Stevens Center for Innovation in Finance

Full-Stack App Developer

Philadelphia, PA

Jun 2024 – Jun 2025

- Enhanced key features of a student loan application to accurately project college costs, using PostgreSQL for data management and Spring Boot for API development
- Designed a responsive [website](#) for the Finiverse project, and created web features for a data visualization platform.

Root2Success

Co-President

Philadelphia, PA

Feb 2022 - Present

- Tutored over 50 hours and developed comprehensive curricula to teach mathematics to over 250 students across more than 25 partner schools in the School District of Philadelphia.
- Led a team of over 80 high school volunteers and organized in-person competitions to promote engagement and foster a love for mathematics.
- Redesigned and developed the [website](#) using React and Node.js

Carnegie Mellon University

Cyber-Security Researcher

Remote

Feb 2025 – August 2025

- Worked with professor [Hanan Hibshi](#) on: The Practical Challenge of Writing Secure Code
- Contributed to a research project on cybersecurity, focusing on using Self-Adapting Language Models (SEAL) for improving email phishing filters.
- Developed a research proposal and methodology for creating an AI-powered filter that can autonomously learn from new phishing tactics.

University of Washington

Machine Learning Researcher

Remote

June 2024 – Aug 2024

- Trained ML models such as XGBoost, LightGBM, CatBoost, Random Forest, K-Means Clustering, and GNNs on an emerging alternative (SMBJ) to traditional PCR techniques used in genetic sequencing and identification.
- Built an AI [system](#) to classify COVID-19 variants from Single Molecule Break Junction (SMBJ) conductance signatures; achieved up to 99.97% accuracy (CatBoost).
- Developed a pipeline: filtering, R^2 -based QC, conductance conversion, and 1D/2D histogram features.

EXTRACURRICULAR ACTIVITIES

FIRST Robotics

Lead Programmer | Mentor | Inspection Lead | Tournament Volunteer | FRC Co-Founder

Philadelphia, PA
Sep 2022 – Present

- Lead Programmer of teams: 22284 and 22279 (state-qualifier) simultaneously
- Co-founded FRC team 10918 Bluesteel Dragons, first year at Masterman since 2019
- Raised \$20k in funding to support new FRC team
- Created autonomous programs utilizing Virtual Field Mapping, Computer Vision, and Pure Pursuit algorithms
- Volunteered at FIRST Lego League (FLL) tournaments and mentored my school's FLL robotics team weekly
- Built an attendance system and [website](#) using an NFC reader and React
- Winner of: RoboJawn 2024 | Motivate Award | Think Award

USChess

Masterman Chess Team Co-Captain | Tournament Director | Chess Camp Instructor

Onsite
Sep 2018 – Present

- Co-Captain for the Elementary/Middle/High School Masterman Chess team
- Held weekly coaching sessions for the Elementary and Middle School
- Certified tournament director | Volunteered to direct the Elementary/Middle School Chess League
- Earned: >\$3,500 | 2x @ World Open (4th) | 3x @ Nationals (5th) | Liberty Bell (1st) | Chess Congress (4th)

PROJECTS

Company Case Competition Analysis | *Figma, Business Strategy*

Jul 2025 – Aug 2025

- Developed a comprehensive business strategy and pitch for Merck, recommending a partnership to create a complementary product for the newly acquired drug, Ohtuvayre.
- Designed and built a prototype for a smart inhaler and tracking app using Figma, visualizing the proposed product and user experience.
- Presented the pitch in a 15-minute presentation for the Leadership in the Business World case competition.

Real-time Edge Derivative Instance Segmentation | *Python, OpenCV, PyTorch, YOLOv11* May 2025

- Authored a mathematical paper on image derivatives (Sobel, Laplacian) and implemented OpenCV edge detectors with code examples.
- Built a real-time object detection and instance segmentation pipeline using YOLOv8 with FPS overlay and COCO class labels.

The Woodlands Cemetery Project | *Web Development, Research*

Mar 2025 – May 2025

- Researched and unearthed the life of James Hepburn Campbell, a 19th-century individual, as part of a rigorous research project for the A.P. United States History course.
- Independently created a professional website in just 3 days to detail Campbell's life and document the team's research process.
- Contributed over 280 hours to the project, demonstrating strong commitment and research skills.

AWARDS

Good Citizen Award

Mar 2025

Awarded by The Union League of Philadelphia for being 2/5,000 nominated by a Philly nonprofit for demonstrating hard work, honesty, curiosity, and loyalty

1st Place - Innovation League 2024

Dec 2024

Awarded \$5000 | Prototyped, developed, and pitched an innovative tech [startup](#) to a panel of judges.

2nd Place - Ctrl+Shift 2024

May 2024

Awarded \$2500 | Developed an innovative [website](#) that redefines AI education.

1st Place- AMC 12B with Distinction and AIME Qualifier

Nov 2024

Scored 1st in my school among seniors and juniors 2 years in a row and scored in the top 5% of all test takers in the country, qualifying me for the American Invitational Mathematics Examination

3rd Place- Penn Mathematics Contest

April 2025

Awarded by The University of Pennsylvania for winning 3rd Place, scoring a 65/80 at the Penn Math Contest (PMC IV)

Questbridge National College Match Finalist and Questbridge Prep Scholar

Sep 2025

Recognized for academic excellence, personal qualities, and for being a competitive candidate for their college partners.

Congressional App Challenge Finalist

Nov 2024

Recognized by the House of Representatives and Madeleine Dean

Gates Scholarship Semi-Finalist

Dec 2025